LUIZ HENRIQUE MACHADO

Software Engineering Student | Systems Development | Backend & APIs

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GitHub · Portfolio · LinkedIn

PROFESSIONAL SUMMARY

Software Engineering student with professional experience developing and maintaining systems, as well as automating and integrating solutions in a corporate environment. Hands-on experience with backend development, REST APIs, databases and web/mobile applications, with projects built for real problems and deployed to production. Also experienced in requirements gathering, data modeling, version control and technical documentation. Looking for an opportunity in software development to keep growing technically and to help build real systems and solutions.

WORK EXPERIENCE

CNH Industrial Mar 2024 – Aug 2025
Business Intelligence Intern | Curitiba, Brazil

- Developed automation solutions with Python, SQL and corporate tools to reduce manual work and improve internal processes.
- Gathered requirements with stakeholders and translated business needs into specifications and rules for solution development.
- Modeled data, defined business rules and wrote technical documentation for corporate solutions.
- Integrated systems through APIs and tools such as Power Apps, Power Automate and SharePoint.
- Wrote Python scripts to automate routines and process data.
- Used Git for version control and technical documentation of solutions.

FixFacil Assistência Virtual Nov 2023 – Feb 2024
Software Development Intern | Curitiba, Brazil

- Developed and maintained systems running in production.
- Implemented and maintained features of an inventory control system, including its database integration.
- Fixed bugs, refactored code and evolved existing features.
- Contributed to frontend improvements and maintained the system according to business needs.
- Used Git and generative AI tools to support development, debugging and code analysis.

SOFTWARE PROJECTS

Eletrofrio — Smart Alarm Notification System
Eletrofrio Hackathon · Python · Gemini API · Supabase · Docker · Evolution API

- Built an automated pipeline to capture, enrich and process alarms from refrigeration systems, with state control to prevent duplicates.
- Integrated telemetry with the Google Gemini API to analyze temperature curves and generate diagnostics automatically.
- Implemented notifications and a conversational WhatsApp chatbot using the Evolution API, PostgreSQL and context stored in Supabase.
- Containerized the system with Docker Compose and prepared its deployment on Oracle Cloud, including a monitoring dashboard, health checks and backup routines.

Simoldes — Industrial Management System
Full-Stack Web Application · TypeScript · Node.js · MongoDB

- Built a full-stack web application to manage industrial machines, mold programs and operation history.
- Developed REST APIs to query and manage machines, programs, operation history and authentication records.
- Implemented authentication and login/logout event logging for user traceability.
- Built and integrated the frontend, backend and database.
- Deployed the application on Vercel and Render, with version control in Git.

Museu na Mão — Mobile App
Mobile App · React Native · APIs · SQLite

- Developed a mobile app with authentication, geolocation and integration with external APIs.
- Implemented favorites, user profile, search and filters.
- Persisted data locally with SQLite.
- Integrated the Google Places API and the Wikipedia API.

3D Printer Management Dashboard
Node.js · WebSockets · Node-RED · ESP32

- Built a system to monitor 3D printers in real time.
- Implemented WebSocket communication to continuously receive metrics from ESP32 sensors.
- Developed logic to automatically detect offline equipment and anomalies in the metrics.
- Built a web dashboard to track KPIs and equipment status.

EDUCATION

UniSenai PR Mar 2023 – Expected Dec 2026
B.S. in Software Engineering | Curitiba, Brazil

- Relevant coursework: Software Engineering, Software Architecture, Database Systems, Cloud Computing, Automation and Data Analysis.
- Developed software projects for external companies as part of the program.
- Published 2 academic papers on Digital Transformation and Software Quality.

TECHNICAL SKILLS

Languages: TypeScript · JavaScript · Python · SQL

Backend & APIs: Node.js · Express · FastAPI · REST APIs · Evolution API

Frontend: React · React Native · HTML5 · CSS3

Databases: PostgreSQL · Supabase · Redis · MongoDB · MySQL · MariaDB · SQLite

Software Engineering: Object-Oriented Programming (OOP) · Requirements Gathering · Data Modeling · Layered Architecture · API Design · Technical Documentation

Tools: Git · GitHub · Postman · Docker · Docker Compose

Cloud & Automation: Oracle Cloud · Vercel · Render · Power Automate · Power Apps · Google Gemini API

LANGUAGES

Portuguese: Native · **English:** B2/C1 — Fluent · **Spanish:** A2 — Basic

COURSES & CERTIFICATIONS

Power BI Training: From Zero to PRO | Nexos Educação | 27h | 2025
Power BI Training: Advanced Modules | Nexos Educação | 10h | 2025
Python Programming Language – Basic | Fundação Bradesco – Escola Virtual | 18h | 2026
Implementing Databases | Fundação Bradesco – Escola Virtual | 15h | 2026
Data Modeling | Fundação Bradesco – Escola Virtual | 8h | 2023
Introduction to Computer Networks | Fundação Bradesco – Escola Virtual | 15h | 2023